Machine Learning
Homework 2

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1. Let $\{(x^{(i)}, y^{(i)})\}_{i=1}^{N}$ be N training samples, with $x^{(i)} \in \mathbb{R}^d$ and $y^{(i)} \in \{-1, +1\}$. Here N and d are positive integers. Consider the ℓ_2 soft margin SVM:

$$\begin{aligned} &\min_{w,b,\xi} \frac{1}{2}\|w\|_2^2 + \frac{C}{2}\sum_{i=1}^{N}\xi_i^2 \\ &\text{s.t.} \quad y^{(i)}(w^T x^{(i)} + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad i = 1, \ldots, N. \end{aligned} \tag{1}$$

The variables are $w \in \mathbb{R}^d$, $b \in \mathbb{R}$, and $\xi = (\xi_1, \ldots, \xi_N)^T \in \mathbb{R}^N$; $C > 0$ is given and $\|\cdot\|_2$ is the Euclidean norm.

(a) For fixed w and b, find the minimizing slack ξ_i for each sample. Substitute your result into the objective to obtain an unconstrained problem in w and b.

(b) The constraints $\xi_i \geq 0$ can be omitted without changing the optimal value. For this equivalent formulation, write the Lagrangian $\mathcal{L}(w, b, \xi, \alpha)$, where $\alpha = (\alpha_1, \ldots, \alpha_N)^T \in \mathbb{R}^N$ collects the nonnegative margin-constraint multipliers α_i.

(c) Set the derivatives of $\mathcal{L}$ with respect to w, b, and each ξ_i equal to zero, and give the resulting relations.

(d) Derive the dual problem, specifying its objective and all constraints on α.

Ans:

(a)

(b)

(c)

(d)

2. For a positive integer n and variable $x = (x_1, \ldots, x_n)^T \in \mathbb{R}^n$, consider

$$\begin{aligned} \min_x \; & -\sum_{i=1}^{n} \log(a_i + x_i) + \sum_{i=1}^{n} \beta_i x_i \\ \text{s.t.} \quad & x_i \geq 0 \; (i = 1, \ldots, n), \qquad \sum_{i=1}^{n} x_i = B. \end{aligned} \tag{2}$$

The parameter vectors $a = (a_1, \ldots, a_n)^T$ and $\beta = (\beta_1, \ldots, \beta_n)^T$ are given, with $a_i > 0$, $\beta_i \in \mathbb{R}$, and $B > 0$. The logarithm is natural.

(a) Write the Lagrangian and KKT conditions. Define all multipliers.

(b) Let x^* be the optimal solution and μ^* the multiplier of $\sum_{i=1}^{n} x_i = B$ at the optimum. Express x_i^* in terms of μ^* and give an equation determining μ^*. Then find x^* for $n = 3$, $a = (1, 2, 1)^T$, $\beta = (0, 0, 1)^T$, and $B = 2$.

Ans:

(a)

(b)

3. For input vectors $x, z \in \mathbb{R}^2$, consider three SVM kernels: linear $K_{\text{lin}}(x, z) = x^T z$, quadratic $K_{\text{quad}}(x, z) = (x^T z + 1)^2$, and Gaussian $K_{\text{RBF}}(x, z) = e^{-\|x-z\|_2^2/(2\sigma^2)}$. Here $\|\cdot\|_2$ is the Euclidean norm and $\sigma > 0$ is the bandwidth. Give brief explanations; no derivations are required.

(a) Which kernel is the simplest choice for linearly separable classes? Why?

(b) Can a linear kernel represent a circular boundary? Suggest a suitable nonlinear candidate and explain. Several choices may be valid.

(c) A Gaussian-kernel SVM produces many small bends around training samples. To make each sample's influence less local, should σ be increased or decreased? Keep other settings fixed and explain.

Ans:

(a)

(b)

(c)